

Roof Lifecycle Cost Analysis

Laurel market
October 2, 2026

Planned maintenance is projected to save the owner of a typical Laurel school **\$1.29M** over the next 20 years.

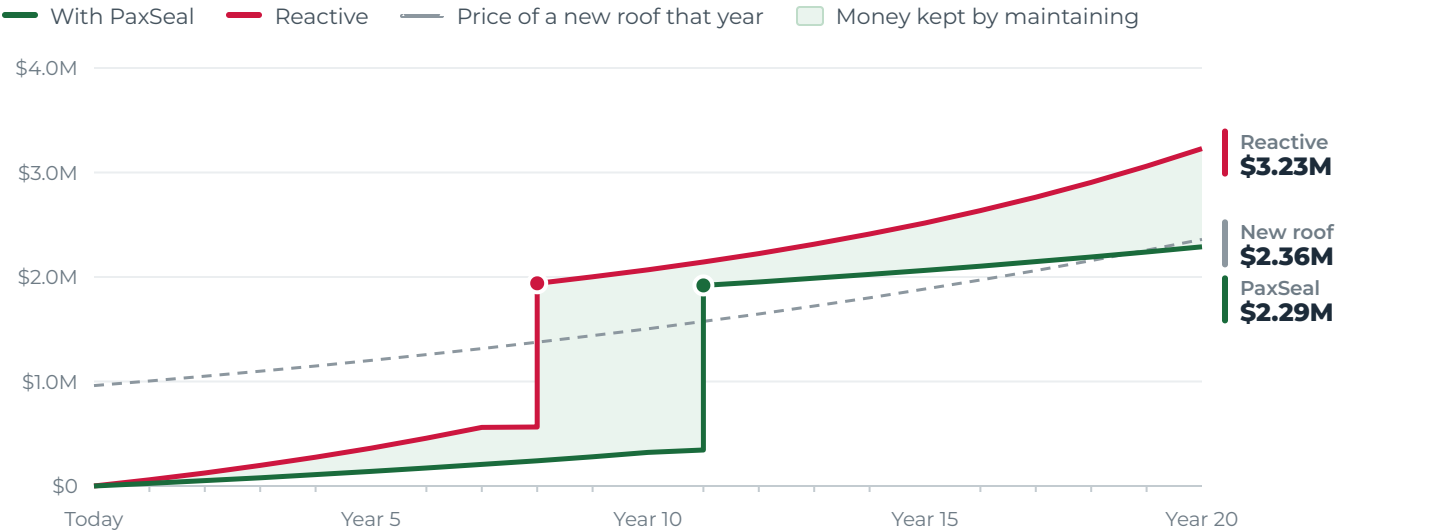
That is 35.1% less than fixing problems only after they leak. Under conservative to upside assumptions, the savings range from \$643K to \$1.7M. In today's dollars the estimate is worth \$674K.

MARKET	PROPERTY	ROOF AREA	SYSTEM	AGE	NEW ROOF TODAY
Laurel	K-12	120,000 sq ft	Mod bit	8 yrs Good	\$960K

PAXSEAL NET COST \$2.39M \$2.29M spent plus roof wear, net of roof life left	REACTIVE NET COST \$3.68M \$3.23M spent plus roof wear, net of roof life left	EMERGENCY COSTS CUT \$1.58M Unbudgeted repairs and storm damage avoided	NEW ROOF PRICE 2× in 15 yrs \$960K today, \$2.36M by year 20
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Cumulative roof spending, next 20 years

Nominal dollars. Vertical jumps are full roof replacements.



Roof life

When each scenario needs a new roof.



Where the savings come from

20-year totals

	PAXSEAL	REACTIVE	SAVINGS
PaxSeal program fees	\$441K	—	-\$441K
Repairs paid by owner	\$234K	\$1.79M	+\$1.56M
Expected storm damage	\$40K	\$60K	+\$20K
Roof replacements	\$1.57M	\$1.38M	-\$199K
Current roof value used	\$576K	\$576K	—
Credit for roof life left	-\$480K	-\$128K	+\$352K
Net 20-year cost	\$2.39M	\$3.68M	+\$1.29M

What this means

- 1 **A later replacement.** The next new roof moves from year 8 to year 11, when it costs less in today's dollars.
- 2 **Smaller repairs.** Fixes made on scheduled visits avoid emergency work at about 3.9× the cost in a school, saving \$1.56M.
- 3 **A predictable budget line.** Emergency repairs become a fixed annual cost the board can plan for, and leaks stop closing classrooms.
- 4 **The program pays for itself.** Each \$1 of fees avoids about \$3.93 of other roof costs.

Local risk profile

Prince George's County, MD (24033)

Thunderstorm wind / derecho, snow load

~3.5 major regional roof-damaging events per 20 years (2010 snow, 2012 derecho, 2016 blizzard). Expected storm damage to this roof over 20 years is **\$40K** if maintained and **\$60K** if neglected. A documented maintenance history also helps when an insurer questions a claim.

Hail	Moderate	
Strong wind	High	
Tornado	High	
Hurricane	Moderate	
Winter weather	Very High	
Ice storm	Moderate	
FEMA National Risk Index rating		

Key assumptions

The basis for each figure in this analysis.

ASSUMPTION	VALUE	BASIS
Roof life	20 / 15 yrs	Maintained / reactive. Fannie Mae service-life tables; +5 yrs per NRC Canada and warranty terms
Roof cost growth	4.6%/yr	BLS Producer Price Index, nonresidential roofing contractors, 2007–2025
Replacement cost	\$8.00/sq ft	Laurel contractor price ranges
Reactive repair cost	3× planned × 1.3	Industry estimates; no primary study found. ×1.3 for leak damage in a school
Storm damage	0.13%/yr	Of roof cost, before insurance. FEMA National Risk Index expected building loss, roof-related hazards
PaxSeal fee	\$16,400/yr	PAX program estimate for this roof size. Your proposal will state the actual fee

Estimates, not a quote. Projections from Laurel market research and industry data, in nominal dollars. The PaxSeal fee is an estimate; your proposal states the actual price. Net cost includes wear on the current roof, less roof life left at year 20. Storm damage is before insurance.